ODD Protocol (ODD 2020): Sections Explained

The **ODD protocol** (*Overview, Design Concepts, and Details*) is a standardized framework for describing agent-based and simulation models. Its goal is to improve model transparency, reproducibility, and communication among researchers.

The protocol is organized into three main sections:a

1. Overview
2. Design Concepts
3. DetailsP

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1. Overview

This section describes what the model is, what it does, and which entities it contains.

1.1 Purpose and Patterns

Purpose

Describes the scientific objectives and motivations of the model.

Questions typically addressed:

- Why was the model developed?
- Which hypotheses are being tested?
- What scientific questions are being investigated?

Example

> The model investigates whether permafrost warming can trigger critical transitions in ecosystem carbon cycling.

Patterns

Describes empirical or theoretical patterns that the model reproduces or seeks to explain.

Examples include:

- Temporal trends
- Spatial patterns
- Statistical relationships
- Experimental observations

Example

> The model reproduces observed increases in ecosystem respiration under warming conditions.

1.2 Entities, State Variables, and Scales

Entities

All objects represented in the model.

Examples:

- Landscape cells
- Individual plants
- Trees
- Herbivores
- Households

State Variables

Variables that characterize each entity.

Entity	State Variables
Cell	Temperature, soil moisture, nutrients
Plant	Biomass, age, species, carbon storage

Scales

Defines the model resolution.

Temporal scale

- Daily
- Monthly
- Annual

Spatial scale

- Individual-based
- Grid cells
- Landscapes

Organizational scale

- Individuals
- Populations
- Ecosystems

Example

> The model operates on a 100 × 100 grid and updates annually for 500 years.

1.3 Process Overview and Scheduling

Describes:

- Which processes occur
- The order in which they occur
- Their frequency

Typical annual schedule:

1. Climate update
2. Plant growth
3. Competition
4. Mortality
5. Decomposition
6. Carbon accounting

This section focuses on workflow and timing rather than equations.

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2. Design Concepts

This section describes the conceptual mechanisms that generate model behavior.

ODD 2020 includes eleven design concepts.

2.1 Basic Principles

Describes the theories or concepts underlying the model.

Examples:

- Resource competition theory
- Metabolic theory
- Resilience theory
- Alternative stable states
- Niche theory

Example

> Ecosystem dynamics emerge from interactions between vegetation growth, nutrient cycling, and climate forcing.

2.2 Emergence

Describes system-level properties that arise from lower-level interactions.

Examples:

- Forest dieback
- Carbon source transitions

- Tipping points
- Species coexistence

Emergent properties are not explicitly programmed but result from model processes.

2.3 Adaptation

Describes how agents modify behavior in response to environmental conditions.

Examples:

- Resource allocation changes
- Behavioral adjustments
- Movement decisions

If adaptation is absent, state:

> Adaptation is not represented.

2.4 Objectives

Describes goals pursued by agents.

Examples:

- Maximize growth
- Maximize fitness
- Acquire resources
- Minimize mortality

Many ecological models do not include explicit objectives.

2.5 Learning

Describes whether agents learn from past experience.

Examples:

- Memory-based decisions
- Reinforcement learning
- Evolutionary adaptation

If absent:

> Learning is not represented.

2.6 Prediction

Describes whether agents anticipate future conditions.

Examples:

- Human planning
- Crop management decisions
- Economic forecasting

Most ecological ABMs do not include prediction.

2.7 Sensing

Describes what information agents can perceive.

Examples:

- Temperature
- Resource availability
- Local competition
- Neighbor density

Sensing determines the information available for decision-making.

2.8 Interaction

Describes how entities influence each other.

Examples:

- Competition
- Facilitation
- Predation
- Nutrient sharing

Example

> Plant growth decreases with increasing neighboring biomass due to competition for resources.

2.9 Stochasticity

Describes all random components of the model.

Examples:

- Random mortality
- Climate variability
- Trait initialization
- Seed dispersal

This section should specify:

- Which processes are stochastic

- Probability distributions used
- Why stochasticity is included

2.10 Collectives

Describes groups that behave as a unit.

Examples:

- Herds
- Colonies
- Social groups

If absent:

> Collectives are not represented.

2.11 Observation

Describes the outputs recorded during simulations.

Examples:

- Biomass
- Species richness
- GPP
- Reco
- NEP
- Spatial variance
- Lag-1 autocorrelation (AR(1))
- Interannual variability (IAV)

This section explains how model outputs are linked to analyses and hypotheses.

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3. Details

This section provides the technical implementation of the model.

3.1 Initialization

Describes model starting conditions.

Examples:

- Initial biomass
- Initial nutrient pools
- Initial species distributions

- Initial climate conditions

Example

> Simulations begin from a mature forest equilibrium at a mean annual temperature of -5°C.

3.2 Input Data

Describes external data supplied to the model during simulations.

Examples:

- Climate observations
- Soil maps
- Remote sensing products
- Experimental forcing data

If external data are not used:

> The model operates without external input data.

3.3 Submodels

Provides detailed descriptions of all model processes.

Typically includes:

- Equations
- Algorithms
- Parameters
- Process-specific assumptions

Examples:

Plant Growth

$$GPP = f(T, N, L)$$

where:

- T = temperature
- N = nutrient availability
- L = light availability

Mortality

$$P_{mort} = 1 - e^{-kS}$$

Decomposition

$$Reco = R_0 e^{\beta T}$$

Nutrient Cycling

$$N_{t+1} = N_t + M - D$$

Submodels are usually the most technical and extensive part of an ODD description.

Typical Reviewer Questions and Corresponding ODD Sections

Reviewer Question	**ODD Section**
Why was the model built?	Purpose
What entities are represented?	Entities, State Variables, and Scales
How does the simulation run?	Process Overview and Scheduling
Which ecological theory supports the model?	Basic Principles
Where do emergent dynamics originate?	Emergence
Which processes are random?	Stochasticity
What outputs are analyzed?	Observation
How are the processes implemented mathematically?	Submodels